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Conf-Gen: Conformal Uncertainty Quantification for Generative Models

Extending Formal Risk Guarantees Beyond Supervised Prediction into LLMs, Image Generators, and Agents

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Conformal risk control (CRC) gives formal uncertainty guarantees for supervised predictors—but a diffusion sample or an LLM’s answer has no ground-truth label to calibrate against. **Conf-Gen** relaxes CRC’s assumptions just enough to reach generative systems directly, producing certified guarantees such as “this image is not a memorized copy,” “this dialogue system has asked enough clarifying questions,” and “this agent’s output is correct”—unifying and generalizing prior one-off attempts to bring conformal guarantees to LLMs.

AT A GLANCE

Problem: Conformal risk control assumes labeled prediction pairs; generative and agentic systems have no natural label to calibrate against.

Why it matters: As generative and agentic systems displace classical supervised predictors, the uncertainty-quantification toolkit built for supervised learning risks becoming obsolete.

Method: Relax CRC’s exchangeable-labeled-pair requirement to an exchangeable-generation-instance requirement, calibrating task-specific risk functionals directly on generated outputs.

Result: A single calibration recipe transfers across image generation, dialogue, and agentic tasks without redesign.

Evidence: Demonstrations on non-memorization guarantees for image generators, clarification-sufficiency guarantees for dialogue systems, and correctness guarantees for agent outputs.

The Big Picture

Conformal prediction and its extension, conformal risk control, are the most practically trusted formal uncertainty guarantees in machine learning—but both were built for supervised prediction, where a labeled calibration pair

(x, y) is available. Modern AI progress is increasingly driven by unsupervised generative models: LLMs, image and video generators, and autonomous agents, none of which produce a prediction against a classical ground-truth label. Conf-Gen asks which of CRC’s assumptions are actually load-bearing, and shows that relaxing just the labeled-pair requirement is enough to bring the same finite-sample guarantee machinery to generative tasks.

Why Existing Approaches Fall Short

Prior attempts to bring conformal guarantees to LLMs typically bolted a supervised-style calibration step onto a generative pipeline in a task-specific, one-off way—for instance treating a single decoding step as if it were a classification label. These approaches do not generalize: each new generative task (image synthesis, dialogue, agentic tool use) required its own bespoke reformulation, with no shared theoretical foundation guaranteeing correctness across tasks.

Core Mechanism

Conf-Gen replaces the labeled-pair requirement with task-specific risk functionals defined directly over generated outputs—distance to the training set as a memorization score, a count of clarifying questions asked, or a verifier’s judgment as a correctness score. Calibration proceeds exactly as in CRC: find the smallest threshold for which the empirical risk, corrected for finite-sample slack, does not exceed the target level. What changes is the object being calibrated: a single generative process’s own outputs, rather than a labeled prediction task.

Technical Formulation

Let $R(\lambda) = \mathbb{E}[\ell(\lambda, X)]$ be the population risk as a function of threshold λ , for a monotone bounded loss ℓ . Calibration selects

$$\hat{\lambda} = \inf \left\{ \lambda : \hat{R}_n(\lambda) + c_n(\delta) \leq \alpha \right\},$$

where $\hat{R}_n$ is the empirical risk over n calibration instances and $c_n(\delta)$ is a finite-sample correction term ensuring $R(\hat{\lambda}) \leq \alpha$ with probability at least $1 - \delta$. Conf-Gen’s relaxation replaces the requirement that (X, Y) be an exchangeable labeled pair with the weaker requirement that the sequence of *generation instances* (and their associated risk scores) be exchangeable—sufficient to preserve the guarantee while accommodating memorization scores, clarification-sufficiency scores, and agent-correctness scores as the risk functional.

Learning or Inference Procedure

1. Define a task-specific, monotone risk functional over generated outputs (memorization distance, clarification-

- tion count, verifier score).
2. Collect an exchangeable sequence of generation instances and compute the empirical risk at a grid of thresholds λ .
3. Calibrate $\hat{\lambda}$ as the smallest threshold whose finite-sample-corrected empirical risk meets the target α .
4. At deployment, apply the calibrated threshold to new generation instances, inheriting the risk guarantee.

What the Guarantee Says

For any monotone bounded risk functional and any exchangeable sequence of generation instances, $R(\hat{\lambda}) \leq \alpha$ holds with probability at least $1 - \delta$ in finite samples—matching the guarantee strength of classical CRC, despite the absence of a conventional labeled calibration set.

Experimental Findings

The paper demonstrates Conf-Gen across three qualitatively different generative settings: certifying that image generator outputs are not memorized copies of training data, certifying that a conversational AI has asked sufficient clarifying questions before committing to an answer, and certifying the correctness of an AI agent’s output—showing the same calibration recipe transfers across modalities and task types without redesign.

Ablations and Interpretation

Unification: Conf-Gen recovers prior ad hoc LLM conformal methods as special cases, showing they were instances of a single underlying relaxation rather than unrelated tricks.

Generality: The framework extends to tasks with no natural notion of a held-out label at all, such as certifying properties of an agent’s multi-step trajectory.

Practical takeaway: Task designers need only supply a monotone risk functional over generation instances; the calibration and guarantee machinery is otherwise unchanged from classical CRC.

Reference

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